



WALL-E AI Platform

Super Master Guide — Kel Shi bMalaf Wahad

Blueprint + Corrections + BOM + Wiring + Firmware + Troubleshooting + STL — Muhad Kamel w Musahhah

Consolidated min: Master Blueprint · Corrected Guide (x3) · Structural Sketch · K-system Manual · Chat additions (PCA9685, Boost Converter, INA219, STL blanks)

Fihres (Table of Contents)

- 1. Muqaddima — Wein Tebda2?
- 2. Qararat Lazim Tak-hodon Qabel Ma Tebda2 (Phase 0)
- 3. El-BOM el-Nihaii el-Kamel + el-Taklife
- 4. Pinout el-Nihaii el-Musahhah (XIAO ESP32S3 Sense)
- 5. Power Architecture w Wiring Rules
- 6. Khatawat el-Bina2 — Marhale be Marhale (Phase 1 → 5)
- 7. Mashekel Mumkin Tewejhak w Hulouha (Troubleshooting)
- 8. El-Code el-Kamel (Firmware Sections)
- 9. 3D Printing / STL Guidance
- 10. Checklist Nihaii — Kel Shi Sar Jahez

1. Muqaddima — Wein Tebda2?

Heda el-malaf howe el-nuskha el-nihaiye elli bteje3 kel el-ma3loumat min kel el-malafat (Master Blueprint, Corrected Guide, Structural Sketch, K-system manual) + kel el-tashih w el-ezafat elli sar ne7kihon bel-chat (PCA9685, Boost Converter, INA219, STL blanks). Iza fi ay ta3arod been malaf w tene, hon mawjoud el-nuskha el-sa77, mesh el-2adime.

Wein tebda2 be-3ajale (suggested first move)

- Yom 1 — bala masare:** 2ra2 seksyon 2 (el-qararat) w 7edded: battery wa7de wala tinten, switch, DRV8833 wala TB6612FNG. Hayde qararat lezim ta5odon 2abel ma te7lob ay 2ata3.
- Yom 1-2:** e7lob el-BOM el-kamel (seksyon 3) min ECA.ir / Roboeq.ir — 3-5 ayyam she7en.
- Bel-wa2et elli el-2ata3 3am toşal:** jahhez el-STL files (seksyon 9) w eb2a jahez lal-7baa3a.
- Lamma toşal el-2ata3:** ebda2 Phase 1 (7baa3a) → Phase 2 (mecanique) → Phase 3 (kahraba) → Phase 4 (barmaje) — be-hal tarteeb, mesh gheiro.

Muhem: la-trekkeb kel el-kahraba 2abel ma teste5dem el-breadboard test (Phase 3, khatwe 11 taht) — kel 7ele lahalha 2abel el-tarkib el-nihaii jowa el-chassis. Heda bi wafferlak wa2et w laham iza fi ghalta.

2. Qararat Lazim Tak-hodon Qabel Ma Tebda2 (Phase 0)

Hal qararat bta2thir 3al BOM, el-taklife, w el-wazn — lezim tetakhad 2abel ma teşref masare.

Qarar	Khiyarat	Ta2thir	Tawsiye
Battery	1× 18650 (3500mAh) aw 2× 18650 parallel (7000mAh)	Runtime: ~3.2 sa3a vs ~6.4 sa3a. Wazn w 3ard el-qa3ide bikaffo la-tinten.	2× 18650 parallel — hamesh amen a3la lal-current, runtime aṭwal, farq se3er bsit (~\$5 ezafi)
Motor Driver	DRV8833 (asli) aw TB6612FNG (badil)	TB6612FNG: ka2a2a a3la, ḥarara a2al, muqawame dakhiliyye a2al = mawtoor a2wa shwaye	DRV8833 mnih w rakhis — TB6612FNG iza el-mizaniyye btesma7 (+\$1-2)
Power Switch	bala switch (lazim tefşol el-battery yadawiyyan kel marra) aw SPDT slide switch	bala switch = mukhaṭara qaşar da2ira w mesh 3amali	Ḑif SPDT switch — \$0.5 bas, w bi7mik el-battery
PTC Fuse	bala fuse aw PTC resettable fuse (500mA-1A)	Ḥimaye basiṭe men qaşar da2ira athna2 el-lahem	Ḑif — \$0.3 bas, ra7iṣa w mufide

3. El-BOM el-Nihaii el-Kamel + el-Taklife

Heda el-BOM el-muwahħad — kel el-ta3dilat (+PCA9685, +Boost Converter, +Capacitor, +Switch, +Fuse, –SD Module el-khareji) mudmaje.

Zata3	Specs	Taklife (\$)	Mulahaza
XIAO ESP32S3 Sense	Mic + Camera + PSRAM 8MB	~\$12	El-3a2el (brain) — PSRAM lezim yenfa3el bel-Arduino IDE
TFT Display	1.5" SPI ST7789 240×240	~\$5	Ta2akkad VCC range bel-datasheet (3.3-5V)
Servos	2× SG53/SG90	~\$4	Pan + Tilt
Track Drive	2× N20 Gear Motors + DRV8833	~\$6	aw TB6612FNG badil (+\$1-2)
Arms NEW	2× SG90 micro-servo + brass wire/PLA arm structure	~\$5	Đaruri — kanou madhkourin bel-image prompt bas bala spec handasi. Shouf Seksyon 4.1
Audio	MAX98357A I2S Amp + 3W Speaker	~\$4	Ta2akkad impedance (4Ω/8Ω) 2abel el-shira2
Distance Sensor	VL53L0X ToF	~\$3.5	I2C — nafs bus ma3 PCA9685 (3unwan mukhtalef)
Touch Button	TTP223	~\$1	⚠ ghazziya men 3.3V bas — mesh men 5V pin
Power System	1-2× 18650 + TP4056	~\$5-10	Hasab 3adad el-batariyyet elli Zarrarto
Wireless Charger	5V Rx + Tx coil kit	~\$3.5	Ekhtiyari
Expansion Sensors	AHT20 (temp/humidity) + MPU6050 (gyro)	~\$3.5	Ekhtiyari — I2C
PCA9685 NEW	16-ch I2C PWM driver	~\$2	Đaruri — bye7el mushkilet el-pin budget w el-motor direction
Boost Converter NEW	MT3608 (2A rated) aw TP4056+Boost combo	~\$1	Đaruri — bysahhe7 khatt el-5V (battery el-kham bteřla3 3.0-4.2V mesh 5V thabet)
Capacitor NEW	100-470μF electrolytic	~\$0.3	Byemna3 brownout reset lal-ESP32 wa2et 7arakat el-mawtoor
Power Switch NEW	SPDT slide	~\$0.5	Tashghil/eřfa2 bala fakk aslak
PTC Fuse NEW	500mA-1A resettable	~\$0.3	Ĥimaye qařar da2ira
SD Module khareji	MicroSD-SPI-Module	~\$2-5	✖ El-Sense fiyo microSD slot mabniyye — mesh mehtaje wa7de khariji

TOTAL (battery wehde): ~\$55-57

TOTAL (2 batariyyet): ~\$60-62

Se3ar be-Toman: ġrob be-mu3addal el-řarf el-yawmi (taqriban 68,000-70,000 Toman lal-Dollar bħasab el-suq el-ħor — te2akkad men se3r el-yom 2abel el-řalab).

Wein teshtri (Mashhad/Iran): ECA.ir w Roboeq.ir bighařto فريمان kel el-2ata3, be-she7en Tipax/Post la-Mashhad, 3-5 ayyam.

4. Pinout el-Nihaii el-Musahhah (XIAO ESP32S3 Sense)

Hayda el-pinout el-wahid el-ša77 — kel el-servos w el-motors šaro 3al PCA9685 (I2C) mesh mubasharatan 3al GPIO, la-ye7el mushkilet el-pin budget (11/11 pins mustaghalle sa77).

Pin 3al Board	Mawsoul 3ala	Mulahaza
5V	Boost Converter OUT (5V thabet)	Main power — mesh mubasharatan men TP4056
GND	Common Ground bus	Kel el-GND lazim yeltaqo sawa 100%
D0 (SDA)	PCA9685 (SDA) + VL53L0X ToF (SDA) + I2C Expansion Port	Mushtarak — kel wehde 3unwan mukhtalef
D1 (SCL)	PCA9685 (SCL) + VL53L0X ToF (SCL) + I2C Expansion Port	Nafs el-bus
D3	TTP223 Touch Sensor	Digital input — pin mustaqel
D4	TFT Display (CS)	Chip Select
D5	TFT Display (DC/RS)	Data/Command
D6 / MOSI	TFT Display (MOSI)	⚠ boot pin — iza sar mushkile b-el-2ela3, jarreb tbaddel
D7 / SCK	TFT Display (SCK)	⚠ boot pin
D8	MAX98357A (BCLK)	I2S
D9	MAX98357A (LRC)	I2S
D10	MAX98357A (DIN)	I2S
Onboard SD slot	mabniyye bel-board	Este3milo mubasharatan — mesh mehtaj wasel yadawi
Onboard Mic+Camera	mabniyye bel-board	Lal-Sentry Mode w el-Realtime Voice

Qanawat PCA9685 (mawsoul 3ala D0/D1)

PCA9685 Channel	Mawsoul 3ala
CH0	Pan Servo (SG53)
CH1	Tilt Servo (SG53)
CH2	DRV8833 AIN1 (Left motor dir A)
CH3	DRV8833 AIN2 (Left motor dir B)
CH4	DRV8833 BIN1 (Right motor dir A)
CH5	DRV8833 BIN2 (Right motor dir B)
CH6 	Left Arm Servo (SG90)
CH7 	Right Arm Servo (SG90)

DRV8833 VM (motor power) byakhod men Boost Converter OUT (5V) mubasharatan — mesh men XIAO wala men 3.3V. El-arm servos (CH6/CH7) byakhdo 5V men PCA9685 V+ — nafs rail el-Pan/Tilt servos, mesh mehtaj pin GPIO jdid abadan (fadhi 8 channels ba3ed 3al PCA9685 la-tawse3at mustaqbaliyye).

4.1 — El-Arms: Spec Handasi Kamel (kan na2es)

El-arms kanou madhkourin bas bel-"AI Image Generation Prompt" (description sathiyye la-sowar) — bala ab3ad, servo, wala BOM entry. Hayda el-spec el-nihaii lezim yenfaz.

3unsur	Spec
Nou3 el-Servo	2× SG90 micro-servo (nafs nou3 el-Pan/Tilt, sghar w khafaf — mesh SG53 el-akbar)
Tarkib	Mthabbat 3al janbein el-trapezoidal body, b-irtifa3 taqriban nuşf el-jism (~2.5cm men foo2 el-qa3ide)
Ḥarakat el-Arm	Single-axis rotation (foo2/taḥt aw amam/wara) — mesh multi-joint, la-yebqa basit w khafif
Tool/Horn	Brass wire (1mm, nafs el-material el-skeletal frame el-madhkour bel-Blueprint el-asli) aw dhira3 PLA maṭbou3 — ʔoul taqriban 3-3.5cm
PCA9685 Channel	CH6 (shmel), CH7 (yamin)
Folтийet	5V — nafs rail el-Pan/Tilt (PCA9685 V+), mesh men XIAO 3.3V
STL	Mesh mawjoud b-el-blanks el-jahzin (Seksyon 9) — lezim ta3mel mount bracket sghir 3al trapezoidal_body.stl b-Tinkercad (2 faṭḥat servo horn 3al janbein)

I2C Sensors Ezafiyye (nafs el-bus D0/D1)

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Sensor	Wazife	3unwan (Address)	Mulahaza
AHT20	Harara/Ruṭoube el-ghorfe	0x38	Bala ta3aroḍ
MPU6050	Ḥes el-hazz/el-2alb — "Dizzy Animation"	0x68	Bala ta3aroḍ
BH1750	Ḍaw2 el-ghorfe — "noum" tlqa2i	0x23	Bala ta3aroḍ, ~\$1
INA219	Qiyas daqiq lal-battery % w current	0x40 aw 0x41	⚠ nafs 3unwan PCA9685 — lazim tbaddel wahad
TCS34725	Ta3rif el-alwan	0x29	⚠ nafs 3unwan VL53L0X ToF — lazim XSHUT logic

5. Power Architecture w Wiring Rules

El-Mushkile el-Ahamm: Khatt el-5V

TP4056 el-3adi howe charger + protection **bas** — mesh boost converter. El-OUT+/OUT- byetla3 bass foltiyet el-battery el-kham (3.0V-4.2V), mesh 5V thabet. Hayda kan bi-khalli el-XIAO (5V pin), el-servos, w el-shashe ya5edo foltiyet mutaghayyira — w hayda byezid ihtimal brownout.

El-ḥal: ḍif Boost Converter Module (MT3608, ~\$1, rated 2A) been el-battery/switch w baqi el-a2ta3, bas ya5rej 5V thabet. Aw dawwer 3ala "TP4056 + Boost 5V combo module" jahez.

Tawzi3 el-Ḥmol (la-akthar kafa2a)

El-Ḥmol	Maṣdar	Sabab
Servos, XIAO, TFT, MAX98357A	5V ba3d el-Boost	Ḥassasin lal-foltiyet, byeḥtajo esteqrar
DRV8833 (VM) — mawtoorat bas	Battery el-kham mubasharatan (3.0-4.2V)	DRV8833 byet7ammal 2.7-10.8V, w el-mawtoorat ma byeḥtajo Boost — bitwaffer ḥmol 3an el-converter

Qawa3ed el-Tawsil

- Battery(at) → TP4056 (B+/B-)
- Wireless Receiver → TP4056 (IN+/IN-) [iza mawjoud]
- TP4056 OUT+ → Boost Converter IN+ → **5V bus wa7de** tewṣal la: XIAO (5V pin), Servos (3an ṭariq PCA9685 V+), MAX98357A (VIN)
- Battery el-kham (mubasharatan, qabel el-Boost) → DRV8833 (VM)
- GND wa7de** tewṣal la-kel shi bala esteth-na2: XIAO, PCA9685, DRV8833, TFT, Amp, ToF, Touch, Boost Converter
- Capacitor (100-470μF) been 5V w GND, qarib men el-servos

Iza ma fi Common GND kamel: el-servos bysiru yrofto (jitter) w ya3mlo random noise. Hayda a2wa sabab shay3 lal-mashekel el-kahrabaiyye.

6. Khatawat el-Bina2 — Marhale be Marhale

Phase 0 — Qararat (yom wahad, bala masare)

Shouf Seksyon 2 fouq — 7edded el-battery, el-driver, el-switch, w el-fuse 2abel ma tetlob.

Phase 1 — 3D Printing (usbou3)

- 3addel malaf el-Structural Sketch: wassa3 3ard el-ra2be men 2.1cm la **2.8-3.0cm** (la-yes3 el-SG53/SG90 Pan Servo bi-raḥa)
- Material: PETG aw PLA+, Infill 20-25% Gyroid, Layer height 0.12-0.16mm
- Eṭba3: Head box (6.0×3.5×4.5cm), Neck platform (2.9cm), Trapezoidal body (qa3ide 6.3cm), Base/tracks (6.3×6.5×3.2cm)
- E7req 4-5 thoqoub sghar (2-3mm) taḥt el-shashe la-ṣot el-speaker
- Sanding (P240→P400) → Base coat aṣfar matṭ → Weathering (acrylic burnt umber/dark grey, dry-brush la-rust w scratches)

Phase 2 — Tajmi3 Mikaniki (2-3 ayyam)

- Zero el-servos (Home Calibration: `servo.write(90)`) 2abel ma terkabon bel-ra2be, w thabbet el-horns be-zawiye 90°
- Thabbet 2× N20 Motors b-qa3r el-robot (Silicone/Hot Glue), w waṣṣel el-jenzeer/tracks
- Thabbet 2× SG90 arm servos 3al janbein el-trapezoidal body** (~2.5cm men foo2 el-qa3ide), w waṣṣel el-brass wire/PLA arm 3al horn
- Thabbet Wireless Receiver Coil b-ka3eb el-chassis (iza mawjoud)
- Thabbet TP4056 + Boost Converter + el-battery(at) bel-makan el-mukhaṣṣaṣ

Phase 3 — Kahraba (Wiring) — 3ala Breadboard Awwalan, Mesh Direct Chassis

- Waṣṣel el-buss el-5V w el-GND common (shouf Seksyon 5)
- Hoṭṭ el-capacitor (100-470µF) been 5V w GND, qarib men el-servos
- Waṣṣel PCA9685 3al I2C (D0/D1)
- Waṣṣel Pan/Tilt servos 3al PCA9685 (CH0/CH1)
- Waṣṣel DRV8833 3al PCA9685 (CH2-CH5) + el-mawtoorat
- Waṣṣel arm servos 3al PCA9685 (CH6/CH7)**
- Waṣṣel TFT (D4/D5/D6/D7)
- Waṣṣel ToF sensor 3al I2C (D0/D1)
- Waṣṣel Touch sensor (D3)
- Waṣṣel MAX98357A + speaker (D8/D9/D10)

Phase 4 — Barmaje (Firmware Bring-up)

- Arduino IDE → Board: `XIA0_ESP32S3 Sense` , PSRAM: Enabled (OPI), Flash: 8MB
- Nazzel el-Libraries: `TFT_eSPI` , `Adafruit_PWMServoDriver` , `Adafruit_VL53L0X` , `WiFiClientSecure` , `WebSocketsClient` , `WiFiManager`
- Flash test sketch el-asasi (Seksyon 8.1) → ta2akkad servos/screen/ToF/motors shaghalin
- Ekhtebar step-by-step: audio → motors → ToF → screen → servos, wa7ad wa7ad 2abel ma tewaṣṣel el-kel sawa

Phase 5 — Milestones Checklist

- ☐ Ṭalab el-2ata3 min ECA/Roboeq
- ☐ Ṭbaa3a w talwin el-chassis (be-3ard ra2be 2.8-3.0cm)
- ☐ Breadboard test: ToF + screen + servos + motors 2abel el-tarkib el-nihaii
- ☐ Laham w tarkib da5el el-chassis 12cm
- ☐ Flash el-code el-kamel + test Web Dashboard

7. Mashekel Mumkin Tewejhak w Hulouha

El-Mushkile	El-Sabab el-Muhtamal	El-Hal
ESP32 Reset / Brownout wa2et ḥarakat el-mawtoor	Peak current draw men Servos/Motors	Ta2akkad el-servos 3al 5V ba3d el-Boost Converter, ḍif capacitor 100-470μF been 5V w GND
TFT Screen bayḍa / ma bi-teshtaghel	SPI pin conflict aw driver settings ghalat b- <code>User_Setup.h</code>	Ta2akkad MOSI(D6), SCK(D7), CS(D4), DC(D5), w ST7789 driver mfa33al
Servos byerfoto / ḥaraka mesh mustaqirra	GND mesh common, aw PWM signal fih noise	Waṣṣel kel el-GND sawa (ESP32, TP4056, Servos, PCA9685)
Audio noise / crackling	Power rail ḍa3if aw I2S clock mismatch	Faṣel aslak el-audio 3an aslak el-motor driver la-tetjannab EMI
ESP32 ma byes3ed tehm̄el kel el-a2ta3 (pin shortage)	11 GPIO bas, el-design el-2adim kan mehtaj 17	Este3mel PCA9685 lal-servos w el-motors (I2C bas)
I2C bus conflict / device mesh mestajib	2 sensors 3ala nafs el-3unwan (mesalan PCA9685 w INA219 kelon 0x40)	Baddel 3unwan wahad 3an ṭariq solder jumper 3al board
SD card mesh 3am teshtaghel	Este3mal module khareji badal el-onboard slot	El-Sense fiyo microSD slot mabniyye — este3milo mubasharatan, la-teḍif module khareji
Boot loop / ESP32 mesh 3am ye2la3	Jehaz mawṣoul 3ala boot pin (D6/D7/D2) w maskeh el-khat wa2et el-tashghil	Jarreb tefṣol el-jehaz el-mawṣoul 3al pin heda wa2et el-2ela3, aw baddel el-pin
Realtime API WebSocket crash	PSRAM mesh mfa33al — audio buffer ma3ando makan	Tools → PSRAM: OPI PSRAM bel-Arduino IDE
Battery bteṭfa2 bsor3a / runtime qaṣir	Battery wehde bas (3500mAh), aw Boost Converter byakhod current akbar men el-output (inefficiency)	Este3mel 2× 18650 parallel — runtime ~6.4 sa3a badal ~3.2
Analog battery gauge mesh ḍabt / mesh mutaḥharrek	PWM mesh momken yeḥarrek ebret analog gauge — heda ghalṭa mafhoumiyye men el-design el-asli	Este3mel INA219 (I2C) la-qiyas daqiq badal el-fikra el-2adime
Arm servo bi-jerr (stall) w byeskhan	SG90 dayfa la-wazn el-arm, aw el-servo waṣel 3ala angle khaṭa2 barra range 0-180°	Ta2akkad SERVO_MIN/SERVO_MAX mu3ayara sa7, w el-arm khafif (brass wire mesh PLA taqil)

8. El-Code el-Kamel (Firmware Sections)

Kel sections lezim yenzammo b-malaf `.ino` wahad 3al ESP32, ma3 el-Libraries el-madhkoure b-Phase 4.

8.1 — Test Sketch el-Asasi (servos + screen + ToF + motors 3al PCA9685)

```

#include <Wire.h>
#include <Adafruit_PWMServoDriver.h>
#include <Adafruit_VL53L0X.h>
#include <SPI.h>
#include <TFT_eSPI.h>

Adafruit_PWMServoDriver pwm = Adafruit_PWMServoDriver(0x40);
#define PAN_CH 0
#define TILT_CH 1
#define L_AIN1 2
#define L_AIN2 3
#define R_BIN1 4
#define R_BIN2 5
#define ARM_L_CH 6 // Left arm servo (SG90)
#define ARM_R_CH 7 // Right arm servo (SG90)
#define SERVO_MIN 150 // pulse count out of 4096, calibrate per servo
#define SERVO_MAX 600

Adafruit_VL53L0X lox = Adafruit_VL53L0X();
TFT_eSPI tft = TFT_eSPI();
#define TOUCH_PIN D3

void setServoAngle(uint8_t channel, int angle) {
    int pulse = map(angle, 0, 180, SERVO_MIN, SERVO_MAX);
    pwm.setPWM(channel, 0, pulse);
}

void driveMotor(bool left, int8_t direction) { // -1, 0, 1
    uint8_t inA = left ? L_AIN1 : R_BIN1;
    uint8_t inB = left ? L_AIN2 : R_BIN2;
    if (direction > 0) { pwm.setPWM(inA, 0, 4096); pwm.setPWM(inB, 0, 0); }
    else if (direction < 0) { pwm.setPWM(inA, 0, 0); pwm.setPWM(inB, 0, 4096); }
    else { pwm.setPWM(inA, 0, 0); pwm.setPWM(inB, 0, 0); }
}

void drawEyes() {
    tft.fillScreen(TFT_BLACK);
    tft.fillCircle(50, 60, 30, TFT_BLUE);
    tft.fillCircle(50, 60, 10, TFT_WHITE);
    tft.fillCircle(130, 60, 30, TFT_BLUE);
    tft.fillCircle(130, 60, 10, TFT_WHITE);
}

void performBounceAnimation() {
    setServoAngle(TILT_CH, 120); delay(80);
    setServoAngle(TILT_CH, 60); delay(120);
    setServoAngle(TILT_CH, 90);
}

// Arms: wave animation ("hi" gesture) – CH6/CH7 3al PCA9685
void performWaveAnimation() {
    setServoAngle(ARM_R_CH, 150); delay(200);
    setServoAngle(ARM_R_CH, 90); delay(150);
    setServoAngle(ARM_R_CH, 150); delay(200);
    setServoAngle(ARM_R_CH, 30); delay(400); // arm balash
}

// Arms: raise both ("excited") – istaghlal wa2t el-Bounce animation
void performArmsUpAnimation() {
    setServoAngle(ARM_L_CH, 150);
    setServoAngle(ARM_R_CH, 150);
    delay(500);
    setServoAngle(ARM_L_CH, 30);
    setServoAngle(ARM_R_CH, 30);
}

void setup() {
    Serial.begin(115200);
    pinMode(TOUCH_PIN, INPUT);
    Wire.begin(D0, D1); // board macros, mesh raw 0/1
    pwm.begin();
    pwm.setPWMFreq(50);
    setServoAngle(PAN_CH, 90);
    setServoAngle(TILT_CH, 90);
    setServoAngle(ARM_L_CH, 30); // arms down = neutral rest position
    setServoAngle(ARM_R_CH, 30);
    tft.init();
    tft.setRotation(1);
    drawEyes();
    if (!lox.begin()) Serial.println("ToF Sensor Init Failed!");
}

void loop() {
    VL53L0X_RangingMeasurementData_t measure;
    lox.rangingTest(&measure, false);
    if (measure.RangeStatus != 4 && measure.RangeMilliMeter < 200) {
        performBounceAnimation();
        delay(1000);
    }
    delay(100);
}

```

8.2 — INA219 Battery Reading (badal el-analog gauge)

```
#include <Adafruit_INA219.h>
Adafruit_INA219 ina219(0x41); // 3unwan mbaddal – mesh 0x40 (PCA9685)

void setup() {
  ina219.begin();
}

float readBatteryPercent() {
  float busVoltage = ina219.getBusVoltage_V();
  float pct = (busVoltage - 3.0) / (4.2 - 3.0) * 100.0;
  return constrain(pct, 0, 100);
}
```

8.3 — OpenAI Realtime API (Live Voice) — Muṣaḥḥaḥ

Ghayyer `model=...` b-akher esem model mawjoud bel-tawthiq el-ḥali men OpenAI wa2t el-tanfiz — mesh esem thabet ktabnah hon.

```
#include <WiFi.h>
#include <WiFiManager.h>
#include <WiFiClientSecure.h>
#include <WebSocketsClient.h>
#include <driver/i2s.h>

WebSocketsClient webSocket;
const char* apiKey = "YOUR_OPENAI_API_KEY"; // ballešha men secrets.h, mesh hardcoded

void setupI2SMic() {
  i2s_config_t i2s_config = {
    .mode = (i2s_mode_t)(I2S_MODE_MASTER | I2S_MODE_RX | I2S_MODE_PDM),
    .sample_rate = 16000,
    .bits_per_sample = I2S_BITS_PER_SAMPLE_16BIT,
    .channel_format = I2S_CHANNEL_FMT_ONLY_LEFT,
    .communication_format = I2S_COMM_FORMAT_STAND_I2S,
    .intr_alloc_flags = ESP_INTR_FLAG_LEVEL1,
    .dma_buf_count = 4,
    .dma_buf_len = 256
  };
  i2s_driver_install(I2S_NUM_0, &i2s_config, 0, NULL);
}

void websocketEvent(WStype_t type, uint8_t * payload, size_t length) {
  switch (type) {
    case WStype_CONNECTED:
      webSocket.sendTXT(String("{\"type\":\"session.update\",\"session\":{\""
        "\"modalities\":[\"text\",\"audio\"],\""
        "\"instructions\":\"You are WALL-E, a witty AI assistant.\"}"));
      break;
    case WStype_BIN:
      i2s_write(I2S_NUM_1, payload, length, NULL, portMAX_DELAY);
      break;
    case WStype_DISCONNECTED:
      Serial.println("[WS] Disconnected – reconnect logic here");
      break;
    default: break;
  }
}

void setupRealtimeVoice() {
  WiFiManager wm;
  wm.autoConnect("WALL-E-Setup");
  webSocket.beginSSL("api.openai.com", 443, "/v1/realtime?model=REPLACE_WITH_CURRENT_MODEL");
  webSocket.onEvent(websocketEvent);
  String headers = "Authorization: Bearer " + String(apiKey) + "\r\nOpenAI-Beta: realtime=v1\r\n";
  webSocket.setExtraHeaders(headers.c_str());
  webSocket.enableHeartbeat(15000, 3000, 2);
}
```

Amane: la-ṥoṭ el-API key bel-code w terfa3o 3a GitHub aw techarko ḥada. Este3mel `secrets.h` munfaṣel w maḥṭoṭ b- `.gitignore` .

8.4 — Voice Memo (Whisper) — Tasjil + Raf3

```
#include <SD.h>
#include <driver/i2s.h>
#include <HTTPClient.h>
#include <WiFiClientSecure.h>

#define REC_SAMPLE_RATE 16000
#define REC_BITS 16
#define REC_CHANNELS 1
File recFile;
bool isRecording = false;

void writeWavHeader(File &f, uint32_t sampleRate, uint16_t bitsPerSample, uint16_t channels) {
  uint32_t byteRate = sampleRate * channels * bitsPerSample / 8;
  uint16_t blockAlign = channels * bitsPerSample / 8;
```

```

uint8_t header[44];
memcpy(header, "RIFF", 4);
uint32_t chunkSize = 0; memcpy(header + 4, &chunkSize, 4);
memcpy(header + 8, "WAVEfmt ", 8);
uint32_t subchunk1 = 16; memcpy(header + 16, &subchunk1, 4);
uint16_t audioFormat = 1; memcpy(header + 20, &audioFormat, 2);
memcpy(header + 22, &channels, 2);
memcpy(header + 24, &sampleRate, 4);
memcpy(header + 28, &byteRate, 4);
memcpy(header + 32, &blockAlign, 2);
memcpy(header + 34, &bitsPerSample, 2);
memcpy(header + 36, "data", 4);
uint32_t subchunk2 = 0; memcpy(header + 40, &subchunk2, 4);
f.write(header, 44);
}

void patchWavSizes(const char* path) {
    File f = SD.open(path, FILE_WRITE);
    if (!f) return;
    uint32_t fileSize = f.size();
    uint32_t dataSize = fileSize - 44;
    uint32_t riffSize = fileSize - 8;
    f.seek(4); f.write((uint8_t*)&riffSize, 4);
    f.seek(40); f.write((uint8_t*)&dataSize, 4);
    f.close();
}

void startRecordingToSD() {
    if (SD.exists("/rec.wav")) SD.remove("/rec.wav");
    recFile = SD.open("/rec.wav", FILE_WRITE);
    if (!recFile) { Serial.println("SD open failed"); return; }
    writeWavHeader(recFile, REC_SAMPLE_RATE, REC_BITS, REC_CHANNELS);
    isRecording = true;
}

void recordingLoopTick() {
    if (!isRecording) return;
    static uint8_t buf[512];
    size_t bytesRead = 0;
    i2s_read(I2S_NUM_0, buf, sizeof(buf), &bytesRead, 20 / portTICK_PERIOD_MS);
    if (bytesRead > 0) recFile.write(buf, bytesRead);
}

void stopRecording() {
    isRecording = false;
    recFile.close();
    patchWavSizes("/rec.wav");
}

String uploadToWhisper(const char* path, const char* apiKey) {
    File f = SD.open(path, FILE_READ);
    if (!f) return "ERROR: file not found";
    WiFiClientSecure client;
    client.setInsecure(); // production: setCACert() ma3 shahadet OpenAI
    if (!client.connect("api.openai.com", 443)) return "ERROR: connect failed";

    String boundary = "---WalleBoundary7MA4YWxk";
    String head = "--" + boundary + "\r\n"
        "Content-Disposition: form-data; name=\"model\"\r\n\r\nwhisper-1\r\n"
        "--" + boundary + "\r\n"
        "Content-Disposition: form-data; name=\"file\"; filename=\"rec.wav\"\r\n\r\n"
        "Content-Type: audio/wav\r\n\r\n";
    String tail = "\r\n--" + boundary + "--\r\n";
    uint32_t contentLength = head.length() + f.size() + tail.length();

    client.printf("POST /v1/audio/transcriptions HTTP/1.1\r\n");
    client.printf("Host: api.openai.com\r\n");
    client.printf("Authorization: Bearer %s\r\n", apiKey);
    client.printf("Content-Type: multipart/form-data; boundary=%s\r\n", boundary.c_str());
    client.printf("Content-Length: %u\r\n", contentLength);
    client.printf("Connection: close\r\n\r\n");
    client.print(head);
    uint8_t buf[512];
    while (f.available()) { size_t n = f.read(buf, sizeof(buf)); client.write(buf, n); }
    client.print(tail);
    f.close();

    String response = "";
    unsigned long timeout = millis();
    while (client.connected() && millis() - timeout < 10000) {
        while (client.available()) { response += (char)client.read(); timeout = millis(); }
    }
    client.stop();
    int bodyStart = response.indexOf("\r\n\r\n");
    return bodyStart >= 0 ? response.substring(bodyStart + 4) : response;
}

void onTouchPress() { startRecordingToSD(); }
void onTouchRelease() {
    stopRecording();
    String transcript = uploadToWhisper("/rec.wav", apiKey);
    Serial.println("Transcript: " + transcript);
}

```

8.5 — Web Dashboard (Local WiFi Control Panel)

```
#include <WebServer.h>
WebServer server(80);

const char HTML_PAGE[] PROGMEM = R"rawliteral(
<!DOCTYPE html><html><head>
<meta name="viewport" content="width=device-width, initial-scale=1">
<style>
body{font-family:Arial;background:#1a1a1a;color:#fff;text-align:center}
.card{background:#2a2a2a;padding:20px;margin:10px;border-radius:10px}
.btn{background:#ffaa00;border:none;padding:10px 20px;font-size:16px;border-radius:5px;cursor:pointer}
</style></head><body>
<h1>WALL-E Control Panel</h1>
<div class="card"><h3>Plugins</h3>
<button class="btn" onclick="fetch('/plugin?type=sentry')">Toggle Sentry Mode</button>
<button class="btn" onclick="fetch('/plugin?type=teleprompter')">Teleprompter Mode</button>
</div>
<div class="card"><h3>Motions</h3>
<button class="btn" onclick="fetch('/bounce')">Bounce</button>
<button class="btn" onclick="fetch('/roll')">Roll / Curiosity</button>
<button class="btn" onclick="fetch('/wave')">Wave (Arms)</button>
</div>
<div class="card"><h3>Status</h3><p id="batt">Battery: --</p></div>
<script>
setInterval(()=>fetch('/battery').then(r=>r.text()).then(t=>document.getElementById('batt').innerText='Battery: '+t+'%'),5000);
</script>
</body></html>
)rawliteral";

void handleBounce() { performBounceAnimation(); server.send(200, "text/plain", "Bounced!"); }
void handleRoll() {
  driveMotor(true, -1); driveMotor(false, 1);
  delay(300);
  driveMotor(true, 0); driveMotor(false, 0);
  server.send(200, "text/plain", "Rolled!");
}
void handleBattery() {
  server.send(200, "text/plain", String((int)readBatteryPercent())); // men INA219 — mesh PWM
}
void handleWave() { performWaveAnimation(); server.send(200, "text/plain", "Waved!"); }
void setupWebServer() {
  server.on("/", []() { server.send(200, "text/html", HTML_PAGE); });
  server.on("/bounce", handleBounce);
  server.on("/roll", handleRoll);
  server.on("/wave", handleWave);
  server.on("/battery", handleBattery);
  server.begin();
}
```

9. 3D Printing / STL Guidance

Fi 4 STL blanks jahzin (men chat sabe2, mershabin bel-2ab3ad el-muṣaḥḥaḥa): `head_box.stl` (60×35×45mm), `neck_platform.stl` (29×20×3mm), `trapezoidal_body.stl` (qa3ide 63mm → foo2 29mm), `base_tracks.stl` (63×65×32mm).

Na2es: arm mount brackets w el-arm segments zetohon mesh mawjoudin bel-blanks el-jahzin. Lezim tzidon yadawiyyan (Seksyon 4.1 fiya el-spec: 2 fathat servo horn 3al janbein el-trapezoidal_body.stl, w dhira3 brass wire/PLA taqriban 3-3.5cm).

Hayde solid boxes be-l-2ab3ad el-khareji el-mazbouṭa bas — el-slicer (Cura/PrusaSlicer) howe yalli byedir el-shell thickness w el-infill. El-cutouts (shashe TFT, 3adaset camera, thoqoub speaker, fathet servo) lazim tzidon b-Tinkercad/Fusion360/Blender aw b-slicer's cut tool.

Khatawat el-Ta3dil (Tinkercad — sahle lal-mubtedi2in)

1. tinkercad.com → hesab majjani → "Create new design"
2. Dbeṭ waḥdet el-qiyas 3ala centimeter
3. Este3mel "Shell" tool la-tfarrigh kel qeṭ3a (byewaffer madde w yekhaffef el-wazn)
4. Dīf fat-ḥat/dawa2er lal-cutouts (shashe, camera, speaker, servo horn)
5. Export → STL, w ehtaḥ b-Cura/PrusaSlicer: Infill 20-25% Gyroid, Layer height 0.12-0.16mm

Badil asra3: dawwer 3ala Thingiverse/Printables 3an "small WALL-E robot head pan tilt" aw "mini robot chassis servo" w 3addel el-qiyasat 3ala el-namouzej el-jahez badal ma tebda2 men el-ṣifr.

10. Checklist Nihaii — Kel Shi Sar Jahez

- ☐ Pinout nihaii muṣaḥḥaḥ w qabel lal-tanfiz fiziya2iyyan (11/11 pins mustaghalle sa77)
- ☐ BOM muḥaddath (+PCA9685, +Boost Converter, +Capacitor, +Switch, +Fuse, −SD module khareji)
- ☐ Qararat Phase 0 metakhda (battery, driver, switch, fuse)
- ☐ Khatawat bina2 men el-ṣifr (ṭbaa3a → mikanik → kahraba → barmaje)
- ☐ Code ekhtebar asasi shaghal bel-PCA9685 (servos/screen/ToF/motors)
- ☐ INA219 battery reading (badal el-fikra el-ghalaṭ lal-analog gauge)
- ☐ Realtime API code muṣaḥḥaḥ (PSRAM + I2S mic + WiFiManager badal hardcoded credentials)
- ☐ Whisper voice memo — tasjil + raf3 kamel
- ☐ Web Dashboard muṣaḥḥaḥ (battery reading ṣaḥi7 + Roll button ḥaqiqi)
- ☐ STL blanks jahzin — cutouts lezim tezado b-CAD tool
- ☐ Arms (spec + BOM + pinout CH6/CH7 + wave animation code) — metakamlin b-hal update

Hayda el-malaf byejma3 kel shi mawjoud la-hal laḥza. Iza zheret ma3loume jdide aw taghyir (mesalan se3r jdid, sensor jdid, aw mushkile jdide waje-htak), khabbirni w n-ḥadith el-nuskha.